

Problem

Each phase load consists of a $20\text{-}\Omega$ resistor and a $10\text{-}\Omega$ inductive reactance. With a line voltage of 480 V rms , calculate the average power taken by the load if:

(a) the three-phase loads are delta-connected

(b) the loads are wye-connected

Step-by-step solution

Step 1 of 6

Consider the value of the impedance Z .

$$Z = 20 + j10\ \Omega$$

Convert to phasor representation.

$$Z = 22.36\angle 26.57^\circ\ \Omega$$

Therefore, the value of the impedance in phasor representation is $22.36\angle 26.57^\circ\ \Omega$.

Step 2 of 6

(a)

Calculate the magnitude value of the line current I_L .

$$I_L = \frac{\sqrt{3}V_L}{|Z|}$$

Substitute $22.36\ \Omega$ for $|Z|$ and 480 V for V_L .

$$\begin{aligned} I_L &= \frac{\sqrt{3}(480)}{22.36} \\ &= \frac{831.38}{22.36} \\ &= 37.18\text{ A} \end{aligned}$$

Therefore, the value of the line current I_L is, 37.18 A .

Step 3 of 6

Calculate the average power P absorbed by the load.

$$P = \sqrt{3}V_L I_L \cos\theta$$

Substitute 37.18 A for I_L , 480 V for V_L and 26.57° for θ .

$$\begin{aligned} P &= \sqrt{3}(480)(37.18)\cos(26.57) \\ &= (30.91\text{ k})(0.8944) \\ &= 27.65\text{ kW} \end{aligned}$$

Therefore, the value of the average power absorbed by delta connected is, 27.51 kW .

Step 4 of 6

(b)

Calculate the value of the phase voltage V_p .

$$V_p = \frac{V_L}{\sqrt{3}}$$

Substitute 480 V for V_L .

$$\begin{aligned} V_p &= \frac{480}{\sqrt{3}} \\ &= 277.13 \text{ V} \end{aligned}$$

Therefore, the value of the phase voltage is, 277.13 V.

Step 5 of 6

Calculate the magnitude value of the phase current I_p .

$$I_p = \frac{V_p}{|Z|}$$

Substitute 22.36 Ω for $|Z|$ and 277.13 V for V_p .

$$\begin{aligned} I_p &= \frac{277.13}{22.36} \\ &= 12.4 \text{ A} \end{aligned}$$

Therefore, the value of the line current I_p is, 12.4 A.

Step 6 of 6

Calculate the average power P absorbed by the load.

$$P = 3V_p I_p \cos \theta$$

Substitute 12.4 A for I_p , 277.13 V for V_L and 26.57° for θ .

$$\begin{aligned} P &= 3(277.13)(12.4)\cos(26.57) \\ &= (10.31 \text{ k})(0.8944) \\ &= 9.22 \text{ kW} \end{aligned}$$

Therefore, the value of the average power absorbed by wye connected is, 9.22 kW.